EFY TESTED S.C. Dwivedi

Multi Switch Control BUZZER/APPLIANCE

SURESH CHANDRA DWIVEDI

Many electronics enthusiasts enjoy designing customised control circuits. Here, we explore an interesting circuit that enables a single buzzer/appliance through multiple switches that may be located in different areas or rooms.

The author's prototype is shown in Fig. 1. As can be seen in the figure, the upper side the breadboard is used for multi-switch control of the buzzer, while its lower side is used for multi-switch control of an appliance.

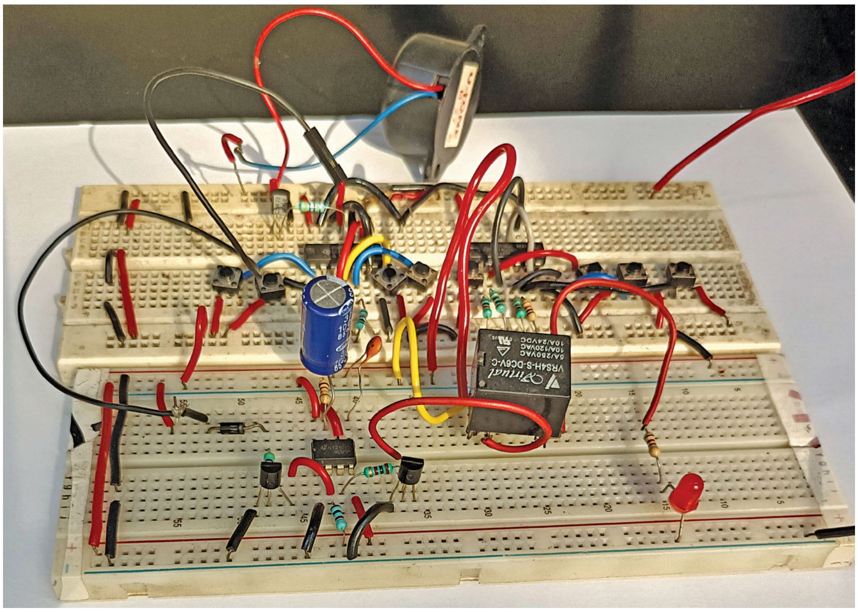

Fig. 1: Author's prototype

Circuit and working

The circuit diagram of the multi-switch control for the buzzer is shown in Fig. 2. It is built around a step-down transformer (X1), bridge rectifier (BR1), 5V voltage regulator 7805 (IC1), two exclusive OR gate 74LS86 ICs (IC2, IC3), buzzer (PZ1), and a few other components.

This circuit utilises two 74LS86 quad 2-input XOR gates to control the buzzer with 8 push-to-on switches (S1 through S8). One end of the eight switches is connected to 5V DC, while the other end is connected to the junctions of one input of an XOR gate and one end of 1-kilohm resistors (R2 through R9). The other ends of the eight resistors are pulled down to ground. Adding pull-down resistors ensures that the inputs of the XOR gates have defined low logic levels when the switches are not pressed.

The outputs of the XOR gates are interconnected in such a way that if any one of the switches is pressed,

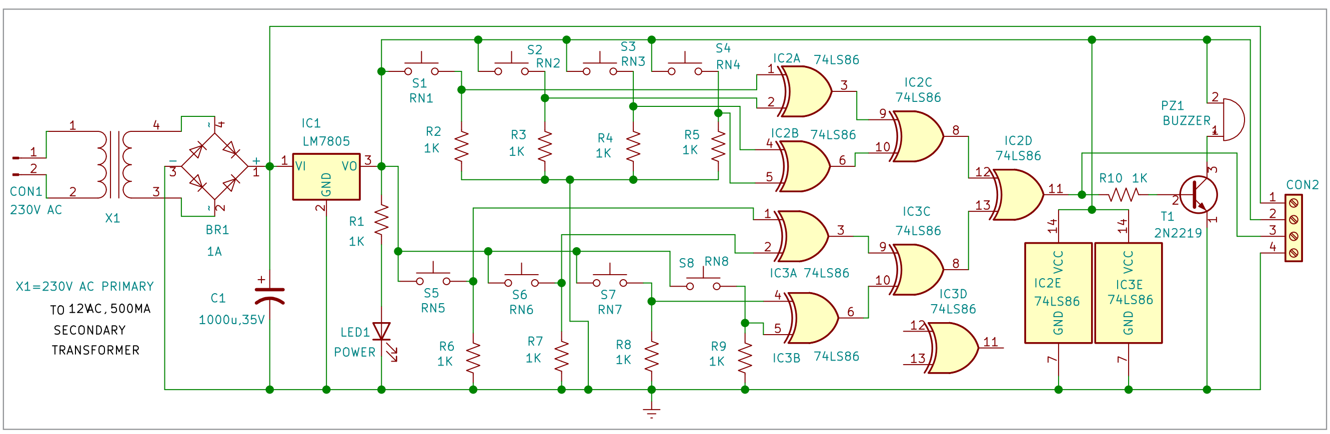

Fig. 2: Circuit diagram of multi-switch control buzzer